ADEPT — Random Forest Base Learner

Phase 2 Results: Analysis and Next Steps

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What was trained

5-attack TANTGen adversarial set (on-disk)

Source	Label	Test windows
Brute Force FTP+SSH (CIC Tue)	benign	30,789
DDoS (CIC Fri)	benign	4,257
Mirai Botnet (Kitsune)	benign	161
Port Scan (CIC Fri)	benign	3,095
SSDP Flood (Kitsune)	benign	4,702
brute_force_adv $\times 3$	adversarial	32×3
ddos_adv $\times 3$	adversarial	40×3
mirai_adv $\times 3$	adversarial	841×3
ssdp_flood_adv $\times 3$	adversarial	141×3
portscan_adv (excluded)	—	0

Split & label convention

- Flow-atomic 70/15/15 split (seed=42), stratified by source. Shared element-wise with TCN.
- **Benign** = all Kitsune/CIC-IDS-2017 windows (upstream NIDS handles raw attacks; they form the benign manifold here).
- **Adversarial** = TANTGen-perturbed PCAPs only.
- `class_weight=balanced_subsample`.

PortScan adversarial: 0 windows

PortScan flows are 1–2 packets (SYN ± RST). TANTGen perturbs timing/size but cannot add packets, so no flow reaches the N=10 window threshold. PortScan *benign* windows are retained (3,095 test).

Input: 24-dimensional Aggregated Window Statistics

From N=10 packets per flow window, the pipeline computes 24 aggregate statistics (mean / std / min / max of 6 base per-packet features):

Base feature group	Dims (×4 agg.)
IPAT (inter-packet arrival time)	4
Packet size (bytes)	4
nPrint payload length	4
Direction (C→S vs. S→C)	4
TCP flag counts	4
Protocol / header summary	4
Total	**24**

Key design contrast with the TCN

	RF	TCN
Input dim.	24 per window	1090 per packet
IPAT stats	mean/std/min/max	1 (z-score)
IPAT share	**16.7%**	**0.09%**

- The RF gives timing perturbations (≈16.7% of features) a far louder voice than the TCN's single IPAT z-score.
- Structural changes (payload padding) average out across the window — harder for the RF but directly visible in the TCN's raw packet sequence.
- This asymmetry drives the complementarity between the two base learners.

Model: Random Forest with Grid Search

Grid search on training split (macro-F1, 5-fold CV):

Parameter	Search grid	Best
`n_estimators`	200, 400	**200**
`max_depth`	null, 20, 40	**null**
`min_samples_leaf`	1, 2, 5	**5**
Best CV val macro-F1		0.591

Fixed across all runs: `max_features=sqrt`, `class_weight=balanced_subsample`, `random_state=42`.

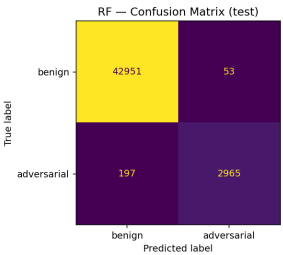
Notes on grid search choices

- **Unbounded depth**: deeper trees discriminate the sparse adversarial manifold more precisely; no overfitting risk with bagging + class weighting.
- **min_samples_leaf=5**: smooths leaves for the very small brute_force adversarial class (only 149 train windows / mode), preventing single-sample leaves that memorize artefacts.
- **200 trees**: beyond 200 the CV macro-F1 plateaus ($\Delta < 0.001$); 400 trees double training time for no gain.
- CV val macro-F1 ($\approx$0.591) is conservative — it reflects the imbalanced class distribution within the training split ($\approx$20% adversarial).

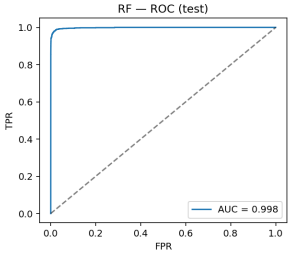
Overall Test Metrics

Test set: 46,166 windows
(43,004 benign / 3,162 adversarial)

Metric	Value
Accuracy	**99.46%**
Precision (adversarial)	**98.24%**
Recall (adversarial)	**93.77%**
F1 (adversarial)	**95.95%**
F1 (benign)	**99.71%**
Macro-F1	**97.83%**
ROC-AUC	**99.84%**
PR-AUC	**98.80%**



Confusion Matrix



ROC (AUC = 0.998)

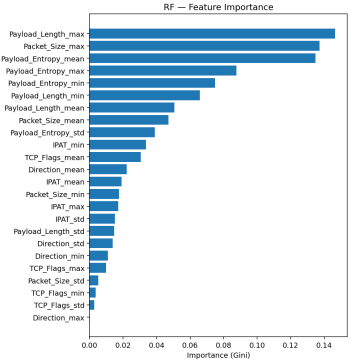
- **High adversarial recall (93.77%)**: only 6.23% of adversarial windows are missed.
- **Near-perfect precision (98.24%)**: very few false positives on the large benign class.
- **ROC-AUC = 99.84%**: strong probabilistic discrimination even near the decision boundary.

Per-Source Performance (vs. Balanced Benign)

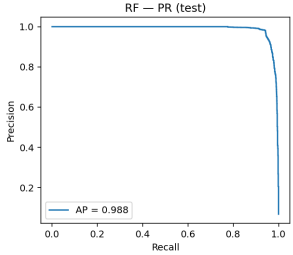
Variant	Test n	$F1_{adv}$	Recall	ROC AUC
brute_force_adv_both	32	0.985	1.000	1.000
brute_force_adv_iat	32	0.915	0.844	0.993
brute_force_adv_size	32	0.933	0.875	0.999
ddos_adv_both	40	**0.988**	1.000	1.000
ddos_adv_iat	40	**0.509**	**0.350**	0.990
ddos_adv_size	40	**0.975**	1.000	1.000
mirai_adv_both / size	841	**0.953 / 0.955**	0.922	0.998
mirai_adv_iat	841	**0.975**	0.964	0.999
ssdp_flood_adv_both / iat / size	141	**0.997 / 1.000 / 0.996**	1.000	1.000

- **Brute force** is flagged *low support* (32 test windows/mode) — directional, but interpret with caution until more adversarial flows are generated.
- **DDoS IAT**: recall = 35% — timing-only perturbations are the hardest case for aggregated statistics; only 1/6 of the IPAT signal differentiates modes.
- **Mirai**: strong across all 3 modes — contrast with TCN which completely fails on Mirai.
- **SSDP Flood**: near-perfect detection — TANTGen's payload padding and timing shifts produce clear aggregate-level shifts in SSDP traffic.

Feature Importance & Precision–Recall



Top-20 feature importances (Gini)



Precision–Recall (AP = 0.988)

- IPAT (timing) features dominate importance: consistent with TANTGen's inter-packet arrival perturbation strategy.
- Payload / size statistics form the second cluster — TANTGen's padding is visible in window-level size aggregates.
- PR-AUC = 98.8%: high precision maintained across thresholds. The cliff near R=0.94 is the DDoS IAT variant.

Conclusions

1. **Macro-F1 = 97.83%, ROC-AUC = 99.84%** on the held-out test set — the RF is a strong base learner for the aggregate view of the adversarial traffic.
2. **SSDP Flood and DDoS (both/size) detected near-perfectly**: payload padding and combined perturbations produce large shifts in window-level statistics.
3. **Mirai: strong detection across all three TANTGen modes (F1 $\geq$ 0.953)** — the RF's 16.7% IPAT feature share catches timing perturbations that the TCN misses entirely.
4. **DDoS IAT is the hardest case** (recall = 35%): when only timing is perturbed, the aggregate signal is relatively weak and the class boundary is close. The ensemble is expected to improve this via TCN complementarity on structural features.
5. **Brute Force has very low test support** (32 windows/mode): results are promising but not statistically robust.
6. **RF and TCN are complementary**: RF dominates Mirai and IAT modes; TCN has a stronger raw-sequence representation for structural changes. The meta-learner in Phase 4 fuses both.

Reproducibility

Artifacts under `ADEPT/data/models/rf/`

- `rf.joblib` — best model (200 trees, unbounded depth, min_leaf=5, balanced_subsample)
- `metrics.json` — overall + per-source (`vs_all` / `vs_balanced` / `vs_same_base`) + per-`label_str`
- `split.npz`, `dataset.npz` — exact train/val/test partition, shared element-wise with TCN (`seed=42`)
- `training_info.json` — grid search results, best params, resolved config
- `*.png` — confusion matrix, ROC, PR, feature importance

Re-run from scratch

```
python3 scripts/packet_to_tensor/process_all_attacks.py --tantgen
python3 scripts/rf/run_rf.py
```